

Qualifying Exam in Microeconomics
Arizona State University
June 2022

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Consider this “Robinson Crusoe” single-consumer, single-firm economy with $L \geq 2$ goods. The firm’s production set Y is

$$Y = \mathbb{R}_-^L = \{y \in \mathbb{R}^L \mid y \leq 0\}.$$

The consumer is endowed with consumption $\omega \in \mathbb{R}_{++}^L$ and owns the firm. The consumer’s preferences on the consumption set $X = \mathbb{R}_+^L$ are represented by a *continuous, strictly increasing* function u on X . In addition, u is *quasiconcave*, but not necessarily *strictly* quasiconcave.

Normalize prices so that they sum to one (but they can be negative) and set $S = \{p \in \mathbb{R}^L \mid \sum_{\ell=1}^L p_\ell = 1\}$.

- (a) Define *precisely* the notions of a Pareto optimum (PO) and a Walrasian equilibrium (WE) *in the context of this simple Crusoe economy*.
- (b) Does a WE necessarily exist? If so, then prove it directly; if not, then provide additional sufficient conditions for one to exist and prove your assertion. (By “directly,” we mean by not just invoking a textbook theorem on the existence of a WE.)
- (c) If there *is* a WE under the original assumptions of the problem, then is the WE *allocation* unique? If so, prove it, if not provide a counterexample or explanation. Is the normalized WE *price* unique? If so, prove it, if not provide a counterexample or explanation.
- (d) Does the *conclusion* of the Second Fundamental Welfare Theorem (SFWT) hold here? Explain. Do the *assumptions* of the SFWT hold here? Explain.
[To clarify, by the SFWT we mean this: if (x^*, y^*) is a Pareto optimal allocation, then there is a price $p^* \in S$ such that (x^*, y^*, p^*) is a price (or Walrasian) *quasi-equilibrium*.]

2. Consider an economy with two states of the world, $s = 1, 2$. There are two (Arrow-Debreu) securities: security s pays 1 dollar in state s , and 0 otherwise: if the consumer buys portfolio (x_1, x_2) then its consumption in state s is x_s , $s = 1, 2$. The unit price of security s is $p_s > 0$ dollars. The consumer is endowed with $w > 0$ units of wealth, so its budget constraint is $p_1 x_1 + p_2 x_2 \leq w$, and its consumption set is $\mathbb{R}_+^2$.

The consumer has subjective expected utility preferences with vN-M utility $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ and beliefs $\pi = (\pi_1, \pi_2)$ about the states. The vN-M utility is *strictly increasing* (but is not necessarily concave).

Consider two different price vectors, $p^t = (p_1^t, p_2^t)$ for $t = 0, 1$ and suppose that the consumer chooses portfolio $x^t = (x_1^t, x_2^t)$ at p^t for $t = 0, 1$. Suppose that

- (i) $p_1^0 > p_2^0$;
- (ii) $x_1^0 > x_2^0$; and
- (iii) $p_1^1 < p_2^1$.

In words, the consumer initially demands more of security 1 than security 2 when security 1's price is higher than 2's. Then the prices change so that the security 1's price is lower than 2's.

- (a) What, if anything, do (i) and (ii) imply about the sign of $\pi_1 - \pi_2$? Explain. (Hint: Consider the portfolio $x' = (x_2^0, x_1^0)$, in which x_2^0 is consumed in state 1 and x_1^0 is consumed in state 2—there is no typo here. What can you say about it at $p = p^0$?)
- (b) Prove that $x_1^1 \geq x_2^1$.

REMARK: The *best* answers use simple direct arguments. Alternatively, if you cannot make any headway with the original statement of the problem, you may add the assumptions that u is concave and C^1 with $u' > 0$; if you take this differentiability approach, then keep firmly in mind that nothing rules out 0 demand for a security. This differentiability approach is unlikely to earn a “high pass” for the question though it can certainly earn a “pass” if done well-enough.

3. A risk averse entrepreneur (agent) wants to start a project which requires an initial investment worth an amount $\kappa > 0$. The entrepreneur has no cash of his own and must raise money from a risk neutral lender (principal). The return of the project is a random variable $\theta \in \{\underline{\theta}, \bar{\theta}\}$, $0 < \underline{\theta} < \bar{\theta}$, and is equal to $\bar{\theta}$ with probability $\pi(e)$ and $\underline{\theta}$ with probability $1 - \pi(e)$, where e is the effort exerted by the entrepreneur, which is unobservable to the lender, and can be either 0 or 1, with $\pi(1) > \pi(0)$. A financial contract consists of repayments $\bar{x}$ and $\underline{x}$ depending on whether the project is successful ($\bar{\theta}$) or not ($\underline{\theta}$). If the entrepreneur obtains income I (which is equal to project return minus the repayment) and exerts effort e , his utility is

$$\frac{1 - e^{-\rho I}}{\rho} - \psi(e),$$

where $\rho > 0$ and ψ is the disutility of effort, with $\psi(1) > \psi(0) = 0$. The reservation utility of the entrepreneur is $\bar{u}$. It is also assumed that the following parametric restriction holds: $\rho \left(\bar{u} + \psi(1) + \frac{1 - \pi(1)}{\pi(1) - \pi(0)} \right) < 1$.

- (a) Set up the lender's optimal contracting problem assuming that she wants to induce the entrepreneur to exert $e = 1$.
- (b) Derive the lender's optimal contract when she induces the entrepreneur to exert $e = 1$ (feel free to use any result from the theory that can simplify your derivations). How do the optimal contract repayments change when $\bar{u}$ changes? Explain.
- (c) Suppose the lender has access to a monitoring technology that makes the entrepreneur's effort observable, and as a consequence the contract can be conditioned directly on effort (instead of on the return). How much would the lender be willing to pay for such a technology (assuming that she wants the agent to exert $e = 1$ both when effort is observable and unobservable)?

4. The market inverse demand for a good is given by $p(Q) = a - Q$ if total output $Q \leq a$, and equal to zero otherwise. Two firms produce and sell the good. Each firm has a constant-returns technology; firm 1's unit cost is k , with $0 < k < a$; firm 2's unit cost is either $k + \varepsilon^+$ or $k + \varepsilon^-$, where $-\varepsilon^- = \varepsilon^+ > 0$ and $0 < k + \varepsilon < a$ for $\varepsilon = \varepsilon^-, \varepsilon^+$. Firm 1 believes that the two types of firm 2 are equally likely, so that 2's *expected* cost is just equal to k . The firms simultaneously and independently choose outputs.
- (a) *Derive* the best reply function for firm 1 and for each type of firm 2 in this incomplete-information Cournot game. *Carefully* justify your steps in this derivation.
 - (b) *Identify* the (unique) Bayesian-Nash equilibrium of the incomplete-information game. Assume here and throughout that $\varepsilon^+ < (a - k)/2$ and $\varepsilon^- > k - a$. (These inequalities imply that firm 1 and both types of firm 2 produce positive quantities in any Bayesian-Nash equilibrium, as well in the *complete*-information games in the next part, a fact that you should feel free to use.)
 - (c) Identify (the unique) Nash equilibrium of the complete-information game in which firm 1's cost is k and firm 2's cost is $k + \varepsilon$ for $\varepsilon \in \{\varepsilon^-, \varepsilon^+\}$.
 - (d) Now consider this Bayesian disclosure game. First nature reveals ε to firm 2, but not to firm 1. Next, before the firms choose outputs, each type of firm 2 can choose to disclose (D) its cost to firm 1, or to clam up (C), that is, say nothing. If a firm discloses, its disclosure is *verifiable* by firm 2, so that firm 1 learns 2's cost in that case directly. If a type of firm 2 clams up, then firm 1 learns nothing *directly* from 2's silence. After firm 1 observes firm 2's choice of C or D, the firms then play a simultaneous-move Bayesian-Cournot game.
 - i. Is there a weak Perfect Bayesian Equilibrium (wPBE) in which both types of firm 2 clam up? Explain. (You can answer this question without further elaborate calculations.)
 - ii. Is there a (pure-strategy) wPBE in which firm 1 fails to learn 2's cost? Explain.